MATHS MANTHAN
• TUITION CLASSES •
e-mail: contact@mathsmanthan.co.in | Website:- mathsmanthan.co.in

**Subject:** Mathematics | **CBSE CLASS 10 – WEEKLY TEST 04 (SOLUTIONS)** | **Time:** 45 Minutes

**Class:** Class X (CBSE) | **Chapter 5: Arithmetic Progressions** | **MM: 20**

**Document Type:** Official Step-by-Step Marking Scheme | **Published for:** mathsmanthan.co.in

**CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:**

1. Full marks require explicit formula declaration, all intermediate algebraic steps, and clearly boxed final answers.
2. In internal choice (**OR**) questions, both options are solved completely with identical step-marking rigor.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 1.5M, 2M]**.

Official Evaluation Key | Maths Manthan Meerut

## SECTION A: Multiple Choice & Assertion-Reason MATHS MANTHAN | (5 × 1 = 5 Marks)

**Q1.** The $4^{\text{th}}$ term from the end of the AP $-11, -8, -5, \ldots, 49$ is: [1]

(A) 37 (B) 40 (C) 43 (D) 58

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

**Correct Answer:** **(B) 40** [0.5 Mark]

**Given:** $a = -11$, $d = -8 - (-11) = 3$, $l = 49$.

**Formula / Theorem Used:** $n^{\text{th}}$ term from end $= l - (n-1)d$

**Justification:** [0.5 Mark]

$4^{\text{th}}$ term from end $= 49 - (4-1)(3) = 49 - 9 = \mathbf{40}$.

**Q2.** The famous mathematician associated with finding the sum of the first 100 natural numbers is: [1]

(A) Pythagoras (B) Newton (C) Gauss (D) Euclid

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

**Correct Answer:** **(C) Gauss** [0.5 Mark]

We recall Carl Friedrich Gauss's symmetric pairing identity ($S = 1 + 2 + \cdots + 100$):

**Formula / Theorem Used:** $S_{100} = \dfrac{100(1+100)}{2} = 50 \times 101 = 5050$

**Justification:** [0.5 Mark]

Gauss observed that pairing extreme terms $(1+100), (2+99), \ldots, (50+51)$ creates 50 identical pairs summing to 101, yielding 5050.

**Q3.** If the first term of an AP is $-5$ and the common difference is 2, then the sum of the first 6 terms is: [1]

(A) 0 (B) 5 (C) 6 (D) 15

**▶ STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

**Correct Answer:** **(A) 0** [0.5 Mark]

**Given:** $a = -5$, $d = 2$, $n = 6$. We apply the sum formula:

**Formula / Theorem Used:** $S_n = \frac{n}{2}[2a + (n-1)d]$

**Justification:** [0.5 Mark]

$$S_6 = \frac{6}{2}[2(-5) + (6-1)(2)] = 3(-10+10) = 3 \times 0 = \mathbf{0}$$

**Q4.** The sum of the first 16 terms of the AP $10, 6, 2, \ldots$ is: [1]

(A) $-320$ (B) 320 (C) $-352$ (D) $-400$

**▶ STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

**Correct Answer:** **(A) $-320$** [0.5 Mark]

**Given:** $a = 10$, $d = 6 - 10 = -4$, $n = 16$.

**Formula / Theorem Used:** $S_n = \frac{n}{2}[2a + (n-1)d]$

**Justification:** [0.5 Mark]

$$S_{16} = \frac{16}{2}[2(10) + 15(-4)] = 8(20-60) = 8(-40) = \mathbf{-320}$$

**Q5.** **Assertion (A):** Three consecutive terms $a, b, c$ are in AP if and only if $2b = a + c$.
**Reason (R):** In an AP, the difference between consecutive terms is equal: $b - a = c - b$. [1]

(A) Both (A) and (R) are true; (R) explains (A). (B) Both are true; (R) does not explain (A).
(B) (A) is true but (R) is false. (D) (A) is false but (R) is true.

**▶ STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

**Correct Answer:** **(A) Both (A) and (R) are true; (R) explains (A).** [0.5 Mark]

We apply the equal-difference condition for three terms:

**Formula / Theorem Used:** $b - a = c - b \iff b + b = a + c \iff 2b = a + c$

**Justification:** [0.5 Mark]

Since consecutive differences in an AP are equal, $b - a = c - b$ holds if and only if $2b = a + c$. Both statements are true and (R) proves (A).

**Q6.** Find the sum of the first 22 terms of the AP $8, 3, -2, \ldots$. **[2]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

**Step 1: Identifying Parameters** [1 Mark]

**Given:** AP is $8, 3, -2, \ldots$. Here $a = 8$, $d = 3 - 8 = -5$, and $n = 22$.

We know the sum formula of $n$ terms of an AP:

**Formula / Theorem Used:** $S_n = \frac{n}{2}[2a + (n-1)d]$

**Step 2: Substituting Values & Computing $S_{22}$** [1 Mark]

$$S_{22} = \frac{22}{2}[2(8) + (22-1)(-5)] = 11[16 + 21(-5)] = 11(16 - 105) = 11(-89) = \boxed{-979}$$

**Q7.** If $S_1, S_2, S_3$ be the sum of $n, 2n, 3n$ terms respectively of an AP, prove that: $S_3 = 3(S_2 - S_1)$. **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

**Given:** $S_1 = S_n$, $S_2 = S_{2n}$, and $S_3 = S_{3n}$ for an AP with first term $a$ and common difference $d$.

We know the sum formula of $k$ terms of an AP:

**Formula / Theorem Used:** $S_k = \frac{k}{2}[2a + (k-1)d]$

**Step 1: Writing Expressions for $S_1, S_2, S_3$** [1 Mark]

$$S_1 = \frac{n}{2}[2a + (n-1)d], \quad S_2 = \frac{2n}{2}[2a + (2n-1)d], \quad S_3 = \frac{3n}{2}[2a + (3n-1)d]$$

**Step 2: Simplifying RHS $= 3(S_2 - S_1)$** [2 Marks]

$$S_2 - S_1 = \frac{n}{2}[2(2a + (2n-1)d) - (2a + (n-1)d)]$$

$$= \frac{n}{2}[4a + (4n-2)d - 2a - (n-1)d] = \frac{n}{2}[2a + (3n-1)d]$$

Multiplying both sides by 3:

$$3(S_2 - S_1) = \frac{3n}{2}[2a + (3n-1)d] = \boxed{S_3 \quad \text{[Hence Proved]}}$$

**Q8.** Find four numbers in AP whose sum is 20 and the sum of whose squares is 120. **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

**Given:** Choose the four consecutive AP terms symmetrically:

**Formula / Theorem Used:** Four numbers in AP: $(a-3d)$, $(a-d)$, $(a+d)$, $(a+3d)$

**Step 1: Finding Central Value $a$ from Sum** [1 Mark]

Sum $= 20 \implies (a-3d) + (a-d) + (a+d) + (a+3d) = 20 \implies 4a = 20 \implies a = 5$.

**Step 2: Using Sum of Squares $= 120$** [1.5 Marks]

$$(5-3d)^2 + (5-d)^2 + (5+d)^2 + (5+3d)^2 = 120$$

Using $(A-B)^2 + (A+B)^2 = 2(A^2 + B^2)$:

$$2(25 + 9d^2) + 2(25 + d^2) = 120 \implies 50 + 18d^2 + 50 + 2d^2 = 120$$

$$100 + 20d^2 = 120 \implies 20d^2 = 20 \implies d^2 = 1 \implies d = \pm 1$$

**Step 3: Listing the Four Numbers** [0.5 Mark]

With $d = 1$: $5 - 3 = 2$, $5 - 1 = 4$, $5 + 1 = 6$, $5 + 3 = 8$. (With $d = -1$: $8, 6, 4, 2$.)

Four numbers in AP: **2**, **4**, **6**, **8** (or **8**, **6**, **4**, **2**)

**Q9.** An AP consists of 37 terms. The sum of the three middlemost terms is 225 and the sum of the last three terms is 429. Find the AP. **[3]**

**▶ STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

**Given:** Number of terms $n = 37$ (odd). Let first term $= a$, common difference $= d$.
We identify middlemost and terminal term indices:

**Formula / Theorem Used:** Middlemost term index $= \dfrac{37+1}{2} = 19^{\text{th}}$ term; last three: $a_{35}, a_{36}, a_{37}$.

**Step 1: Equations from Three Middlemost & Three Last Terms** [1.5 Marks]

Three middlemost terms are $a_{18}, a_{19}, a_{20}$:

$$(a+17d)+(a+18d)+(a+19d) = 225 \implies 3a+54d = 225 \implies a+18d = 75 \quad \text{— (i)}$$

Last three terms are $a_{35}, a_{36}, a_{37}$:

$$(a+34d)+(a+35d)+(a+36d) = 429 \implies 3a+105d = 429 \implies a+35d = 143 \quad \text{— (ii)}$$

**Step 2: Solving for $d$, $a$, and Forming the AP** [1.5 Marks]

Subtracting (i) from (ii): $17d = 143 - 75 = 68 \implies d = \frac{68}{17} = 4$.
Substituting $d = 4$ into (i): $a + 18(4) = 75 \implies a = 75 - 72 = 3$.

Required AP: **3**, **7**, **11**, **15**, …, **147**

MATHS MANTHAN • TUITION CLASSES • OFFICIAL SOLUTIONS

**Q10. CASE STUDY: Construction Site Timber Log Stacking**

At a civil infrastructure construction site, heavy cylindrical timber logs are delivered to support shoring works. To prevent accidental rolling and optimize storage space, the site supervisor orders the logs to be stacked in a triangular profile. A total of 200 logs are stacked such that there are 20 logs in the bottom row, 19 logs in the row directly above it, 18 in the next row, and so on, reducing by 1 log in each successive layer. [4]

(i) Write down the values of the first term $a$ and common difference $d$ for the AP representing the number of logs from bottom to top. [1]

(ii) How many logs are placed in the $8^{\text{th}}$ row from the bottom? [1]

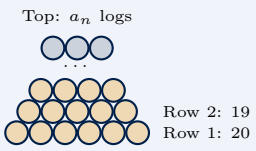
(iii) In how many rows are all 200 logs placed, and how many logs are in the topmost row? [2]

— **OR** —

What is the total combined number of logs placed in the top two rows? [2]

▶ **CASE-BASED SOLUTION & MARKING SCHEME:** [Total: 4 Marks]

**Extracted Scenario Data:** Row-wise log counts from bottom to top form the AP $20, 19, 18, \ldots$ with total sum $S_n = 200$.
We apply the general term and sum formulas of an AP:




**Formula / Theorem**

**Used:** $a_n = a + (n-1)d$ and $S_n = \frac{n}{2}[2a + (n-1)d]$

**Sub-part (i): First Term $a$ and Common Difference $d$** [1 Mark]

First row (bottom) $= 20$, next $= 19$, so $\boxed{\mathbf{a = 20}, \quad \mathbf{d = 19 - 20 = -1}}$.

**Sub-part (ii): Logs in the $8^{\text{th}}$ Row from the Bottom** [1 Mark]

$$a_8 = 20 + (8-1)(-1) = 20 - 7 = \boxed{\mathbf{13}\text{ logs}}$$

**Sub-part (iii): Total Number of Rows ($n$) and Top-Row Logs ($a_n$)** [2 Marks]

$$S_n = \frac{n}{2}[2(20) + (n-1)(-1)] = 200 \implies n(41-n) = 400 \implies n^2 - 41n + 400 = 0$$

Splitting middle term ($-16 - 25 = -41$, $(-16)(-25) = 400$): $(n-16)(n-25) = 0 \implies n = 16$ or $25$. For $n = 25$, $a_{25} = 20 - 24 = -4 < 0$ (impossible; physical log counts cannot be negative). Rejecting $n = 25$:

$$n = \mathbf{16}\text{ rows}, \qquad a_{16} = 20 + 15(-1) = \mathbf{5}\text{ logs in topmost row} \longrightarrow \boxed{\mathbf{16}\text{ rows, }\mathbf{5}\text{ logs}}$$

**Sub-part (iii) [OR Option]: Combined Logs in the Top Two Rows** [2 Marks]

With 16 total rows, the top two rows are Row 15 and Row 16: $a_{15} = 20 + 14(-1) = 6$, $a_{16} = 20 + 15(-1) = 5 \implies$ Combined total $= 6 + 5 = \boxed{\mathbf{11}\text{ logs}}$.

★★★ END OF SOLUTIONS · MATHS MANTHAN MEERUT ★★★